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Unit – 4 : Session – 9 : SLO – 2 | BST Using Linked List — Delete Element (Code)

■ I. deleteElement() — Completed Code

```
• struct node *deleteElement(struct node *tree, int val) {
• struct node *cur, *parent, *suc, *psuc, *ptr;
• if (tree == NULL) {
• printf("\n The tree is empty");
• return tree;
• }
• parent = tree; cur = tree->left;
• while (cur != NULL && val != cur->data) {
• parent = cur;
• cur = (val < cur->data) ? cur->left : cur->right;
• }
• if (cur == NULL) {
• printf("\n The value to be deleted is not present in the tree");
• return tree;
• }
• if (cur->left == NULL)
• ptr = cur->right;
• else if (cur->right == NULL)
• ptr = cur->left;
• else { // Find the in-order successor and its parent
• psuc = cur;
• cur = cur->left;
• suc = cur;
• while (suc->left != NULL) {
• psuc = suc;
• suc = suc->left;
• }
• if (cur == psuc) { // Situation 1
• suc->left = cur->right;
• }
• else { // Situation 2
• suc->left = cur->left;
• psuc->left = suc->right;
• suc->right = cur->right;
• }
• ptr = suc;
```

- }
- if (parent->left == cur)
- parent->left = ptr;
- else
- parent->right = ptr;
- free(cur);
- return tree;
- }